

SCS 1307
Probability & Statistics

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What is a continuous random variable?

- Consider a random variable which can take any value in an interval. Such a random variable is not a discrete random variable because the values in the interval cannot be placed in one-to-one corresponding with counting numbers.

Definition

A random variable which can take any value in an interval is called a **continuous** random variable.

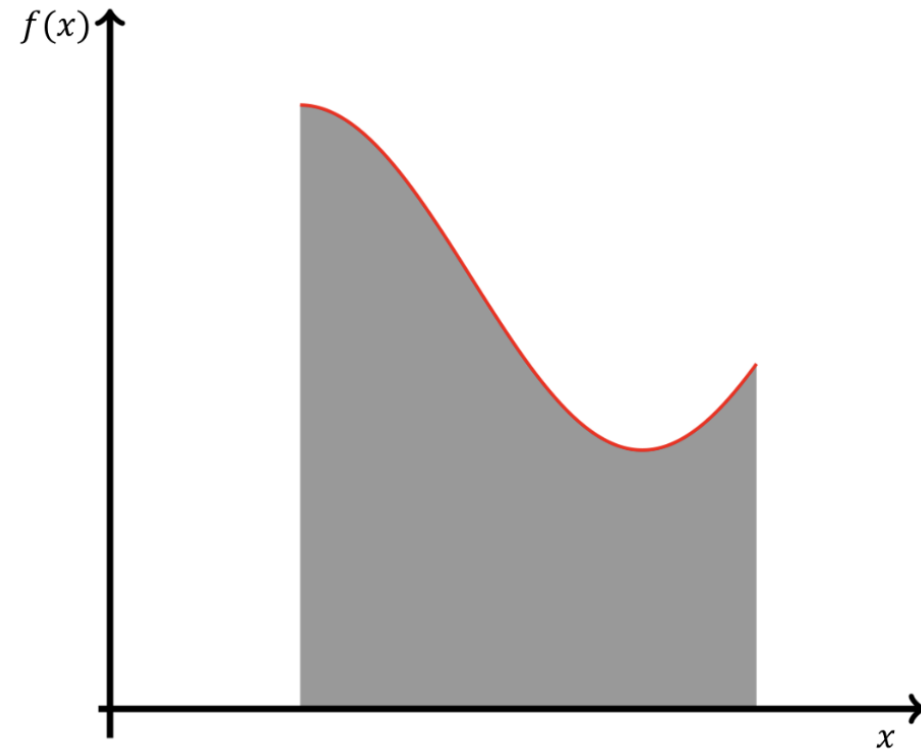
Probability Density Function

The probability properties of a continuous random variable X are specified by its probability density function $f(x)$

This function has the properties that

(1) $f(x) \geq 0$ for every $x \in \mathbb{R}$

(2) $\int_{-\infty}^{\infty} f(x) dx = 1$



The probability of the event $\{a < X \leq b\}$ is found from the density function f as follows

$$P(a < X \leq b) = \int_a^b f(x) dx$$

Note that the density function of a continuous random variable is such that probabilities are given by areas under its graph.

Example

Suppose that the random variable T has a density function given by ,

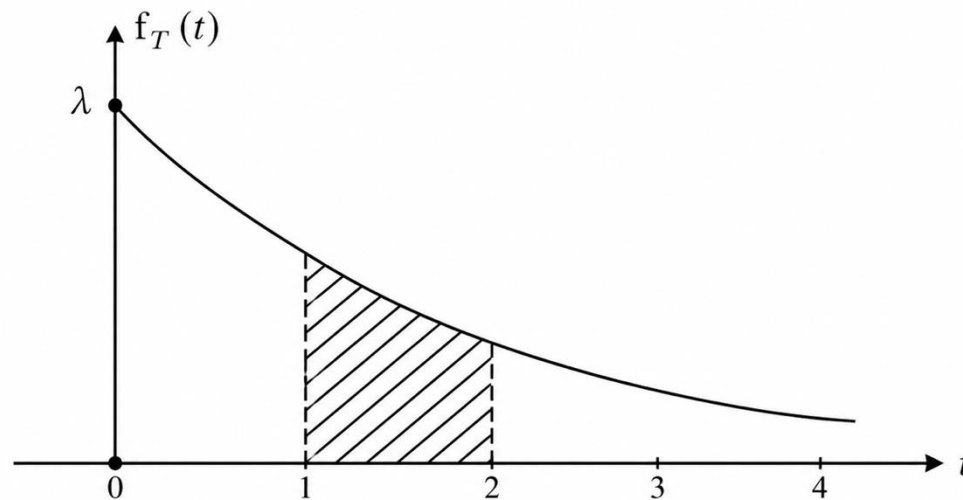
$$\begin{aligned} f_t(T) &= 0 && \text{when } t < 0 \quad \text{and} \\ f_t(T) &= \lambda e^{-\lambda t} && \text{when } t \geq 0 \end{aligned}$$

Show the area corresponding to the $P(1 < T \leq 2)$ in a graph.

Solution

$$\begin{aligned} f_t(T) &= 0 && \text{when } t < 0 \text{ and} \\ f_t(T) &= \lambda e^{-\lambda t} && \text{when } t \geq 0 \end{aligned}$$

$P(1 < T \leq 2)$ is shown below



The area corresponding to

$$P(1 < T \leq 2) = \int_1^2 \lambda e^{-\lambda t} dt = e^{-\lambda} - e^{-2\lambda}$$

has been shaded in the above diagram.

Exercise

A continuous random variable has p.d.f $f(x)$ where

$$f(x) = kx, 0 \leq x \leq 4$$

- (a) Find the value of the constant k
- (b) sketch $y = f(x)$
- (c) find $P(1 \leq X \leq 2.5)$

Solution

A continuous random variable has p.d.f $f(x)$ where

$$f(x) = kx, 0 \leq x \leq 4$$

(a) Find the value of the constant k

Solution

A continuous random variable has p.d.f $f(x)$ where

$$f(x) = kx, 0 \leq x \leq 4$$

(b) sketch $y = f(x) = x/8$

(c) find $P(1 \leq X \leq 2.5)$

Exercise

The continuous r.v X has pdf $f(x)$ where $f(x) = k(4-x)$, $1 \leq X \leq 3$.

- (a) Find the value of constant k
- (b) Sketch $y=f(x)$
- (c) Find $P(1.2 \leq X \leq 2.4)$.

Expectation and Variance

If X is a continuous random variable with p.d.f $f(x)$, then the expectation of X is $E(X)$ where

$$E(X) = \int x f(x) dx$$

$E(X)$ is often denoted by μ and referred to as the mean of X .

The variance of X is $\text{Var}(X)$ where

$$\begin{aligned}\text{Var}(X) &= E(X - \mu)^2 = E(X^2) - \mu^2 \text{ where } \mu = E(X) \\ &= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2\end{aligned}$$

Example

If X is a continuous r.v. with p.d.f $f(x) = \frac{3x^2}{64}$, $0 \leq x \leq 4$, find $E(X)$ and $\text{Var}(X)$.

Solution

If X is a continuous r.v. with p.d.f $f(x) = \frac{3x^2}{64}, 0 \leq x \leq 4,$

$$E(X) =$$

$$\text{Var}(X) =$$